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Ex 10: A* Search Algorithm

CODE:

```
import heapq

def a_star(graph, heuristics, start, goal):
    pq = [(heuristics[start], 0, start, [start])]
    visited = {}

    while pq:
        f, g, current, path = heapq.heappop(pq)
        if current == goal:
            return path, g

        if current in visited and visited[current] <= g:
            continue
        visited[current] = g

        for neighbor, cost in graph.get(current, []):
            new_g = g + cost
            new_f = new_g + heuristics[neighbor]
            heapq.heappush(pq, (new_f, new_g, neighbor, path + [neighbor]))
    return None, float('inf')
```

```
graph = {
   'A': [('B', 1), ('C', 4)],
   'B': [('D', 2), ('E', 5)],
   'C': [('E', 1)],
   'D': [('E', 1)],
   'E': []
}
heuristics = {'A': 7, 'B': 6, 'C': 2, 'D': 1, 'E': 0}
path, cost = a_star(graph, heuristics, 'A', 'E')
print("Shortest Path:", path)
print("Total Cost:", cost)
```

OUTPUT:

```
============== RESTART: C:/Users/Saiko/Documents/CSA1738/EXP-10.py ==========
Shortest Path: ['A', 'C', 'E']
Total Cost: 5
```